



HARDWARE REFERENCE

Altura Rally Computer

Jason Hack · Eli Goethel

The Altura Rally Computer is designed exclusively for motorsport and competitive use. It is not approved for road use under standard vehicle regulations. The authors accept no liability for consequences arising from misuse.

1.0 Hardware Components

Microcontroller — ESP32-2432S028R

The Cheap Yellow Display (CYD) is a compact ESP32-based development board with an integrated 320×240 TFT display and resistive touchscreen, making it well-suited for a self-contained rally instrument.

Parameter	Value
Processor	Dual-core Xtensa LX6
Clock speed	160 MHz (reduced from 240 MHz for power and thermals)
Core 0 role	GPS data processing (background task, priority 2)
Core 1 role	Display and touch (foreground, Arduino loop)
Wireless	WiFi and Bluetooth disabled at boot — reduces RF noise
Inter-core sync	FreeRTOS mutex protecting the shared data structure

Display — ILI9341 TFT, 320×240 pixels

Parameter	Value
Interface	SPI at 55 MHz via TFT_eSPI library
Orientation	Physical portrait; rotated to landscape in firmware
Backlight	GPIO 21, LEDC PWM at 5 kHz / 8-bit — four brightness levels
Rendering	Two DMA-capable draw buffers (~30 KB each) — double-buffered

Touch — XPT2046 Resistive Touchscreen

Parameter	Value
Bus	VSPI — CLK: 25, MISO: 39, MOSI: 32, CS: 33
Mode	Polled every 5 ms — no interrupt line used
Long-press threshold	2000 ms
Calibration	Hardcoded coordinate mapping calibrated to the physical unit

GPS Module — u-blox NEO-6M (GY-GPS6MV2 breakout)

Parameter	Value
UART pins	RX: 22, TX: 27
Operating baud rate	57600 bps (negotiated from 9600 default on every boot)
Update rate	5 Hz — one complete position fix every 200 ms
Active sentence types	GGA and RMC only — all others disabled
Antenna	Internal patch on GY-GPS6MV2 breakout with hardwired LDO bias

Dynamic model	Automotive — optimised for ground vehicle physics
Static hold threshold	40 cm/s — suppresses Doppler noise when stationary
SBAS	Enabled (WAAS in North America)
Minimum satellite elevation	10 degrees — rejects low-horizon multipath satellites
Config persistence	Volatile — reapplied from firmware on every boot

Storage — ESP32 Internal Flash (NVS)

Key	Type	Description
odo	double	Total odometer in miles
utc_offset	char	UTC time offset, -14 to +14 hours
brightness	uint8	Backlight PWM level (64 / 128 / 192 / 255)
theme	uint8	Active colour theme index (0–16)
font_idx	uint8	Active numeric font index (0–7)

All keys reside under the namespace "rally". NVS write-levelling protects against single-point flash corruption. The odometer is saved at 1-mile intervals and immediately on manual reset or factory reset.

2.0 System Architecture

The diagram below shows how the physical components connect to the ESP32 and how data flows between the GPS background process and the display foreground process.

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+-----+
| ESP32-2432S028R (CYD) |
| |
GPS (NEO-6M) -- | UART2 RX:22 TX:27 |
57600 baud | Core 0: GPS task |---- mutex --+
5 Hz position | Parse RMC + GGA | |
| Motion gates | v
| Accumulate distance | +-----+-----+
| | | RallyData struct |
| Core 1: Display + Touch | | totalTrip |
| Update screen @ 100 ms | | intervalTrip |
| Read touch @ 5 ms | | currentSpeed |
| Run animations | | avgSpeed / max |
ILI9341 TFT -- | SPI 55 MHz GPIO 13,14,15 | | hdop / sats |
320x240 | Backlight PWM GPIO 21 | | GPS time |
| | | reject counters |
XPT2046 Touch-- | VSPI CLK:25 MIS0:39 | +-----+
Resistive | MOSI:32 CS:33 |
| |
Internal Flash--| NVS namespace: rally |
+-----+
```

3.0 Software Libraries

The firmware is written in C++ for the Arduino framework on PlatformIO. The following libraries are used.

Library	Version	License	Role
LVGL	9.2.2	MIT	UI framework — screens, widgets, animations, timers
TFT_eSPI (Bodmer)	2.5.x	MIT	ILI9341 display driver with DMA SPI support
XPT2046_Touchscreen	1.4	MIT	Resistive touch controller polling
NeoGPS (SlashDevin)	4.x	GPL-3.0	GPS sentence parsing — atomic RMC+GGA pairs
Arduino Preferences	built-in	LGPL-2.1	NVS key-value persistence
FreeRTOS	built-in	MIT	Dual-core task management and mutex
esp_task_wdt	built-in	Apache-2.0	Hardware watchdog (5 s timeout, both cores)

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The firmware is open-source and distributed under the GNU General Public License v3.0, inherited from the NeoGPS library dependency. Full source code and contribution guidelines are available in the official firmware repository.

End of Altura Rally Computer Hardware Reference — v1.0